1. Clinical Question & Core Hypothesis

Clinical Question

The path suggests loss of epithelial adhesion (CDH1 down-regulation/EMT) co-occurs with immune-regulatory changes (LAG3) and specific genomic contexts (MSI, KRAS), raising the question: why do some tumors with lymphocyte infiltration nonetheless fail to respond to immune checkpoint therapy? 2) I built on the single path linking CDH1 (E-cadherin loss → EMT/invasion) to immune regulation via LAG3 and the reported correlation between LAG3 and MSI/KRAS contexts; CDH1 and LAG3 provide a direct mechanistic axis (physical/architectural immune exclusion + checkpoint-mediated dysfunction). 3) Supporting evidence (Tier 2) includes the association of LAG3 expression with MSI status and KRAS-related signaling that can shape the microenvironment and T cell function — these support testing whether genomic context modulates the CDH1–LAG3 interaction. 4) I chose therapy_response (immune checkpoint therapy response proxy) because LAG3 and MSI are clinically relevant to immunotherapy sensitivity and because resolving why infiltrated tumors fail therapy has immediate translational impact. 5) Deliberate exclusions: I excluded other imaging features (TLS-like aggregates, TSR) to keep the hypothesis focused on TIL density as the observable immune phenotype; KRAS and MSI were relegated to secondary_modifiers rather than core anchors to preserve mechanistic clarity (max two anchors)?

Working Hypothesis

In epithelial tumors with high TIL density, loss of CDH1 drives EMT-associated architectural changes that create immune-excluded microdomains, reducing productive T cell–tumor cell interactions. Concurrently, upregulation of LAG3 on infiltrating T cells reflects an exhausted/dysfunctional state that suppresses cytotoxic activity. The combination of CDH1-low and LAG3-high defines a TIL-rich but functionally immunosuppressive phenotype that predicts poor response to immune checkpoint blockade (anti-PD-1/PD-L1), even relative to CDH1-high/LAG3-low tumors with similar TIL densities. MSI status and KRAS-driven signaling modulate this axis by shaping antigenicity and inflammatory signaling, thereby altering the magnitude of the CDH1–LAG3–TIL interaction and its impact on therapy response. Therefore, the following testable predictions arise: (1) In TIL-high tumors, regions with CDH1 downregulation will co-localize with immune-excluded zones and reduced T cell activation markers; (2) LAG3 expression on T cells will correlate with exhaustion markers (e.g., PD-1, TIM-3) and be higher in CDH1-low tumors; (3) In vitro and ex vivo systems where CDH1 is knocked down or knocked out in epithelial tumor models will show reduced T cell–mediated killing unless LAG3 signaling is blocked; (4) In vivo models and patient cohorts, a combined signature of CDH1-low/LAG3-high among TIL-rich tumors will predict inferior response to ICIs, with the magnitude of the effect modulated by MSI status and KRAS signaling; (5) Therapeutic benefit may be observed with combination strategies targeting both PD-1/PD-L1 and LAG3 specifically in CDH1-low contexts, particularly where MSI KRAS-related signaling amplifies the immunosuppressive milieu.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	CDH1, LAG3
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CDH1: CDH1 down-regulation/EMT) co-occurs with immune-regulatory changes (LAG3) and specific genomic contexts (MSI, KRAS), raising the question: why do some tumors with lymphocyte infiltration nonetheless fail to respond to immune checkpoint therapy? 2) I built on the single path linking CDH1 (E-cadherin loss → EMT/invasion) to immune regulation via LAG3 and the reported correlation between LAG3 and MSI/KRAS contexts; CDH1 and LAG3 provide a direct mechanistic axis (physical/architectural immune exclusion + checkpoint-mediated dysfunction).

LAG3: LAG3) and specific genomic contexts (MSI, KRAS), raising the question: why do some tumors with lymphocyte infiltration nonetheless fail to respond to immune checkpoint therapy? 2) I built on the single path linking CDH1 (E-cadherin loss → EMT/invasion) to immune

regulation via LAG3 and the reported correlation between LAG3 and MSI/KRAS contexts; CDH1 and LAG3 provide a direct mechanistic axis (physical/architectural immune exclusion + checkpoint-mediated dysfunction).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	KRAS	Supporting modifier from knowledge graph
Secondary	MSI_status	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: and supports rational design of combination therapies (e.g., PD-1/PD-L1 blockade with LAG3 inhibition) tailored to CDH1/LAG3 status and contextual modifiers (MSI, KRAS).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- KRAS: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- MSI_status: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.